

Improving emergency service in rural areas: a bi-objective covering location model for EMS systems

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Abstract Emergency medical service (EMS) systems are public services that often provide the first line of response to urgent health care needs within a community. Unfortunately, it has been widely documented that large disparities in access to care exist between rural and urban communities. While rural EMS is provided through a variety of resources (e.g. air ambulances, volunteer corps, etc.), in this paper we focus on ground ambulatory care. In particular our goal is to balance the level of first-response ambulatory service provided to patients in urban and rural areas by locating ambulances at appropriate stations. In traditional covering location models the objective is to maximize demand that can be covered; consequently, these models favor locating ambulances in more densely populated areas, resulting in longer response times for patients in more rural areas. To address the issue of fairness in semi-rural/semi-urban communities, we propose three bi-objective covering location models that directly consider fairness via a secondary objective. Results are discussed and compared which provide a menu of alternatives to policy makers.

Keywords Emergency medical service · Facility location · Bi-objective optimization · Discrete optimization

1 Introduction

Emergency medical service (EMS) systems are especially organized systems that provide emergency medical service to patients within a service area. This service area could be urban, rural, or some combination of the two depending on how the population is distributed in the geographical region. Unfortunately, rural communities often suffer from inadequate

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medical services, including emergency care. Such a problem is compounded with low expectations that emergency care in rural areas will not be as fast and effective as in urban areas (NCSL 2000). EMS systems are typically evaluated according to how they respond to and treat out-of-hospital cardiac arrest patients. In urban areas, EMS systems tend to have the highest known cardiac arrest patient survival rates. In contrast, EMS systems in rural and semi-rural areas have notably lower cardiac arrest patient survival rates¹ (English 2008).

The large disparities in access to care between rural and urban communities has been widely documented. Poor access to care is due not only to lack of resources (service providers, hospitals, and technicians) (Williams et al. 2001) but also because of delays or difficulties in getting medical care (Rawlinson and Crews 2003). Of particular importance are emergency medical services, as these services not only provide a gateway to health care services to many individuals, but are the first level of response to time-sensitive conditions. While most of us living in the US have access to emergency care, we do not have similar access: “there are regional disparities in [emergency department] access, especially by rurality” (Brendan et al. 2009).

EMS care in rural areas may be provided through a variety of resources (e.g. air transportation, volunteer corps, etc.). However, previous reports indicate that while air transportation may sometimes result in shorter arrival times, in many regions ground transportation results in shorter arrival times to care centers, such that regions should be zoned to avoid inappropriate choice of transport (Lerner et al. 1999). Furthermore, air transportation is costly and not available to all EMS departments. Therefore, providing effective ground transportation is crucial. In this paper we focus on ground ambulatory care. In particular our goal is to balance the level of first-response ambulatory service provided to patients in urban and rural areas by locating ambulances at appropriate stations.

An important factor in determining EMS performance is not only the quality of emergency medical care provided but also the timeliness in which care is provided, or *response time* (McGinnis 2004). A rapid response time by EMS can mean the difference between life and death. In urban areas, the most widely used ambulance response time standard is to respond to 90% of calls within 8 minutes and 59 seconds as compared to responding to 90% of calls within 14 minutes and 59 seconds in rural areas (Fitch 2005). In practicality, it may not be possible to meet this standard depending on the geographical area, the EMS resources available, and the location of EMS resources at the time of the call. Response times may be much longer than the standard, especially in rural or remote areas. One way to reduce the response time is to locate ambulances at appropriate station locations. This problem is known as a *covering problem* where the service to customers depends on the distance between the customer and the facility to which the customer is assigned (Daskin 1995). A call is said to be covered if the response time is within the standard; for example, a call responded to within 8 minutes and 59 seconds or less is considered covered, while if the response time is 9 minutes or more the call is considered uncovered.

Most EMS systems locate and use their resources in a way that maximizes the number of patients (or calls) that can be served within a specified time or distance. Often this is translated to number of demand zones that can be covered, where a demand zone is a geographic region with associated call volume. However, when resources are limited, some demand zones may go uncovered. With a single objective that maximizes the number of covered demand zones, these uncovered demand zones tend to be located at the edges of the region. This results in an inequitable use of resources that impacts patient outcomes. Thus, patient

¹ While these trends are true on the aggregate level, the cardiac arrest survival rates vary considerably across individual EMS systems.

survival rates in rural areas are observed to be significantly lower than in urban areas (English 2008). As we will show in Sect. 6, applying a covering location model with the single objective of maximizing the number of covered demand zones to a semi-rural county results in optimal solutions that locate emergency ambulances at stations that leave the majority of rural demand zones uncovered.

We propose three bi-objective models for locating EMS ambulances in order to reduce the disparities in service among rural and urban areas. The first objective is the traditional covering problem objective of maximizing expected coverage while the second objective is aimed at improving service in rural areas. Since there is no universally accepted way to measure fairness in EMS systems (Marsh and Schilling 1994), we propose three alternatives for the second objective as a means to identify how to best evaluate fairness in EMS systems. The three proposed alternatives to be the second objective are: (1) minimize the maximum distance between each uncovered demand zone and its closest opened station, (2) minimize the number of uncovered rural demand zones, and (3) minimize the number of uncovered demand zones (notice that this last objective is not the same as maximizing the number of covered calls, as each zone has a different demand). A detailed discussion of why these particular objectives are chosen and how they are modeled is provided in Sect. 3.1. These three models are formulated as integer programs. Non-dominated solutions to each bi-objective model are generated using the ε -constraint approach.

The key contribution of this paper is that it provides a model for reducing the geographic disparities in service between different demographics; in particular we focus on urban versus rural areas. This paper does not seek to provide the best way to model equity, since a large body of knowledge regarding equity indicates that stakeholders can evaluate equity in eight distinct ways, suggesting that no one single performance measure for equity satisfies them all (Stone 2001). Rather, we examine the tradeoff between efficiency (coverage) and equity when different formulations for equity are used. Thus, the solution to the model provides decision makers with a set of solutions that can be chosen based upon performance measures of interest. Results indicate that the model that minimizes the maximum distance between each uncovered demand zone and its closest opened station as a secondary objective results in solutions that dominate those of the other models, with respect to the average distance (or weighted average distance) between an uncovered zone to its closest station. On the other hand, a model that uses a secondary objective of minimizing the number of uncovered rural demand zones yields a larger solution set, which may be desirable to the decision maker as it provides more options.

2 Literature review and scope

Models for locating EMS resources typically use variations of the maximal covering location problem (MCLP), where facilities are located at existing stations on the network to maximize the demand that can be served in a specified time or distance (e.g. demand that can be covered) Church and ReVelle (1974). The MCLP model does not make any adjustments for busy vehicles. An extension of the MCLP, the Maximal Expected Covering Location Problem (MEXCLP), proposed by Daskin (1983) considers that vehicles may be busy; however this busy probability is exogenous to the model and assumed to be the same for all ambulances. Recognizing that the busy probabilities for each vehicle may differ, other facility location models have been developed. For example, Hogan and ReVelle (1986) consider the issue of backup coverage, or secondary coverage of a demand zone. Backup coverage is required in high-demand areas to maintain a uniform level of service when EMS ambulances

can respond to only one call at a time. Batta et al. (1989) introduced the Adjusted MEXCLP (AMEXCLP) which recognizes that servers do not operate independently and thus may have different busy probabilities which depend on the server location. They develop correction factors for the busy probabilities in an embedded Hypercube model (the hypercube model was first developed by Larson (1974, 1975)) to allow for dependencies between servers and server location. In the model presented in this paper we will use AMEXCLP to calculate expected coverage. Other papers which consider different coverage assumptions (e.g. allowing for partial coverage or restricting coverage) include ReVelle et al. (1996), Berman and Krass (2002), and Karasakal and Karasakal (2004), among others. For a detailed and recent review of ambulance location models see Brotcorne et al. (2003). While differing in how coverage is calculated, all of these papers focus on the single objective of maximizing coverage.

Several papers do consider multiple objectives for ambulance location problems. Daskin et al. (1988) integrated different covering models such as multiple, excess, backup and expected covering models. For example, they reformulate the hierarchical objective set covering problem (Daskin and Stern 1981) into a multi-objective set covering problem, which allows them to derive the trade-off between the number of facilities and the extra coverage. Pirkul and Schilling (1988) modeled the objective of maximizing covered calls while simultaneously considering workload capacities and backup service. Pirkul and Schilling (1991) extended this model with the addition of workload limits on the facilities and the quality of service delivered to the uncovered demand zones. The workload limit refers to the specific amount of demand that can be served by one facility. The workload limit condition is formulated as a constraint to make the model more realistic. The quality of service is modeled as the total distance from uncovered demand zones to the nearest facility, and the resulting model is solved using a solution procedure based on Lagrangian relaxation. Narasimhan et al. (1992) extended the model to consider multiple levels of backup. More recently, Araz et al. (2005) developed a multi-objective covering location model based on previously developed models (Hogan and ReVelle 1986; Pirkul and Schilling 1988). Their model has three objectives: (1) maximizing the population covered by one vehicle, (2) maximizing the population with backup coverage, and (3) minimizing the total distance from locations at a distance bigger than a specified distance standard for all zones. The problem is solved using a fuzzy goal programming approach.

Although there are many extensions to covering location models, there are few models that explicitly address fairness of service to patients in rural areas. Under a single covering objective (e.g., maximizing the expected covered demand), patients in urban areas are generally covered at the expense of rural patients, leading to adverse patient outcomes in rural areas. Even as more EMS ambulances become available, covering models tend to continue to concentrate EMS resources in urban areas. Keeney and Winkler (1985) differentiate between *ex ante equity* (resource allocation decisions that lead to differences in service or health risks) and *ex post equity* (differences in service or health outcomes). In practice, equity is often evaluated *ex post*. Savas (1978) describes several qualitative measures for measuring equity in public services and indicates how each should be selected and used.

Marsh and Schilling (1994) provide a comprehensive review of twenty measures for equity in facility location applications, focusing on *ex post equity*. They focus on comparing the functional forms of the equity rather than analyzing how the functional forms of equity lead to various facility location decisions in real-world examples. They note that there is little or no consensus on the best way to measure equity.

In some EMS facility location models, equity has been indirectly evaluated by the choice of model. For example, one way to ensure fairness in patient outcomes is to add chance or

reliability constraints that enforce minimum levels of service throughout the system (ReVelle and Hogan 1989; Ball and Lin 1993). Another way to indirectly model equity is to use a p -center formulation rather than a p -median formulation for locating p ambulances at stations (Marsh and Schilling 1994). The p -median problem minimizes demand-weighted travel distance whereas the p -center problem minimizes the maximum distance between nodes and their nearest ambulance, thus providing a measure of equity. However, these approaches have not been explicitly evaluated in terms of equity nor have they examined the interaction between equity and efficiency measures (such as maximizing coverage).

In this paper, we propose three objective functions to locate EMS ambulances so as to reduce the disparities in service between rural and urban areas. Each of these objective functions is used in a bi-objective discrete optimization model that evaluates the tradeoffs between coverage and fairness. The first objective function minimizes the maximum distance between uncovered demand zones and opened stations. This objective function assigns the uncovered demand zones to the closest opened stations, in order to minimize the distance from an uncovered zone to an open station. This is important because even if a zone cannot be responded to within the response time standard, patient survivability rates are directly related to response time (or equivalently, distance) (Larsen et al. 1993). The second objective function minimizes the total number of rural demand zones that cannot be covered. This objective function considers the trade-offs between the number of rural demand zones that can be covered and the amount of demands in these demand zones that can be covered. The third objective function minimizes the total number of uncovered demand zones, either urban or rural. The idea of the third objective function is similar to the second objective function, but it does not consider the type of uncovered zones (i.e., urban or rural). The proposed objective functions provide guidelines for locating ambulances, while allowing decision makers to simultaneously improve the quality of service in both rural and urban areas.

3 Covering location model

This section introduces a bi-objective covering location model for locating EMS ambulances at preexisting rescue stations that balances efficiency (i.e., expected coverage) and equity.

3.1 Performance measures for efficiency and equity

Most of the literature surrounding server-to-customer systems (such as EMS systems) have examined how to *efficiently* allocate resources according to various performance measures (Goldberg 2004). Arguably, the most widely used performance measure for efficiency is expected coverage, or the expected number of calls that can be answered in a pre-specified timeframe (Goldberg 2004). Thus, the first objective is to maximize the expected number of requested calls that can be covered, namely, Z_1 (1).

Equity, on the other hand, has been explicitly considered in few operations research models for EMS systems. In EMS facility location models, equity has been indirectly evaluated by the choice of model (see Sect. 2). Erkut et al. (2009) compare different facility location models for EMS systems in terms of efficiency. However, no work, to our knowledge, quantitatively evaluates models for EMS systems in terms of equity, nor do they discuss the best way to model equity. Currently there is no consensus on the best way to measure equity; in fact, the social science literature (Stone 2001) suggests that there are eight ways to define equity depending on the perspective of the stakeholder. Since it is widely documented that large disparities in access to service exist between rural and urban regions (Rawlinson and

Crews 2003) our second objective is to improve fairness by reducing disparity in service among different geographic regions within a community. With this in mind, we propose three alternative objective functions for improving fairness in rural areas:

- minimize the maximum distance between uncovered demand zones and their closest opened stations (Z_{2a} , (2a)),
- minimize the number of uncovered rural demand zones (Z_{2b} , (2b)), and
- minimize the number of uncovered demand zones (Z_{2c} , (2c)).

The first secondary objective (Z_{2a}) is chosen because the notion of coverage (incorporated in the first objective) considers a zone uncovered when the ambulance cannot respond to the zone within a prespecified time standard (e.g. 9 minutes) thus, an ambulance that is 9 min away is not any different than one that is 15 min away in the AMEXCLP model, however, the difference in these response times can have a huge impact on the survival probability of the patient (see for example Larsen et al. 1993). The AMEXCLP favors locating ambulances in more densely populated areas leaving more rural areas uncovered, thus objective Z_{2a} , aims to reduce the proportion of persons who have much delayed service. This objective is similar to the p -center formulation for facility location models, but is only concerned with distances between uncovered demand nodes and their closest facility (as opposed to distance from all nodes to closest facility). Objectives Z_{2b} and Z_{2c} directly address the issue of disparity. These objectives give equal value to all relevant zones, in Z_{2c} all zones are considered equal, while in Z_{2b} we target rural zones since these are usually the ones that go uncovered. Notice that this is not the same as maximizing expected coverage because expected coverage is weighted by the proportion of demand in each zone.

3.2 Formulation

These three alternative objective functions are selected one at a time to be used as a second objective, resulting in three distinct models. We have two types of decision variables which are y_{ik} (a 0–1 variable that indicates if demand zone i is covered by at least k ambulances), and x_j (the number of ambulances at station j). There are three constraints, shown in (3) to (5). The first constraint (3) limits the total number of ambulances available to be located to T . The second constraint (4) limits the maximum number of ambulances that can be located at a single station to S . In the third constraint (5), a demand zone can receive service from a station as long as that station is open (i.e., there is at least one vehicle located there). Equations (6) and (7) represent non-negativity and integrality constraints.

$$\text{Maximize} \quad Z_1 = \sum_{i=1}^n \sum_{k=1}^{l_i} h_i w_k y_{ik} \quad (1)$$

$$\text{Minimize} \quad Z_{2a} = \max_{i \in U} \left\{ \min_{j \in O} (d_{ij}) \right\} \quad (2a)$$

$$\text{Minimize} \quad Z_{2b} = \sum_{i=1}^n (1 - y_{i1}) r_i \quad (2b)$$

$$\text{Minimize} \quad Z_{2c} = \sum_{i=1}^n (1 - y_{i1}) \quad (2c)$$

$$\text{Subject to:} \quad \sum_{j=1}^m x_j \leq T \quad (3)$$

$$x_j \leq S, \quad j = 1, \dots, m \quad (4)$$

$$\sum_{k=1}^{l_i} y_{ik} \leq \sum_{j \in J_i} x_j, \quad i = 1, \dots, n \quad (5)$$

$$x_j \in \{0, 1, \dots, S\} \quad j = 1, \dots, m \quad (6)$$

$$y_{ik} \in \{0, 1\} \quad i = 1, \dots, n; \quad k = 1, \dots, l_i \quad (7)$$

Where the decision variables are:

$$y_{ik} = \begin{cases} 1 & \text{if demand zone } i \text{ is covered by at least } k \text{ ambulances} \\ 0 & \text{otherwise} \end{cases}$$

x_j = the number of ambulances located at station j

The following list summarizes the parameters used:

w_k = the probability that the k th vehicle is available (see below)

h_i = the call volume in demand zone i

d_{ij} = the distance from station j to demand zone i

$$r_i = \begin{cases} 1 & \text{if demand zone } i \text{ is in rural area} \\ 0 & \text{otherwise} \end{cases}$$

T = the total number of ambulances to be located

S = the maximum number of ambulances allowed to be located at each station

$J_i = \{j | d_{ij} \leq D\}$ = set of stations that can cover demand zone i

D = the maximum distance that can be reached within 9 minutes (4 miles)

$U = \{i | y_{i1} = 0\}$ = set of uncovered demand zones

$O = \{j | x_j \geq 1\}$ = set of opened stations

$l_i = \min\{S \times |J_i|, T\}$ (upper bound on the number of vehicles that can cover a demand zone)

n = the number of demand zones

m = the number of stations

Note that a demand zone i can be covered by 1 up to l_i vehicles such that $y_{i1} \geq y_{i2} \geq \dots \geq y_{il_i}$. For example, If demand zone i is covered by 1 vehicle, $y_{i1} = 1$ and $y_{i2} = \dots = y_{il_i} = 0$. If demand zone i is covered by 2 vehicles, $y_{i1} = y_{i2} = 1$, and $y_{i3} = \dots = y_{il_i} = 0$. However, it is not necessary to enforce this using a constraint of the form $y_{i,k} \geq y_{i,k+1}$ because of the definition of w_k given by equation (11). By definition, w_k is the probability that the k th vehicle is available (while $k - 1$ vehicles are busy) such that w_k is greater than w_{k+1} , and since the objective is to maximize Z_1 which weighs each y_{ik} by w_k , then for each demand zone i it will be optimal to let $y_{i,k} \geq y_{i,k+1}$.

In our model, when calculating the expected number of calls that can be covered we account for the fact that, even if ambulances are stationed within the coverage distance, they may be busy and therefore unable to respond to a call. The probability that a randomly selected vehicle will be busy, p , depends on the number of ambulances that are deployed. To estimate p , we use actual data of the system, which is captured by (8), where, λ is the average number of calls per hour, $1/\mu$ is the average service time per call (hours), and T is number of ambulances that are deployed. This definition of p assumes that all ambulances operate independently. This assumption can be relaxed using the correction factor given by Batta et al. (1989) in an embedded hypercube model. The hypercube model Larson (1974, 1975) has several underlying assumptions: (1) calls for service arrive according to a Poisson process, (2) if a call arrives while all servers are busy, it enters at the end of a queue and will be served in a FIFO manner. If there are k ambulances that may respond to a call, the probability that the k th vehicle will be dispatched or is available is calculated from the probability that $k - 1$ ambulances are busy and the k th vehicle is available. The probability that the k th vehicle is available (w_k) is shown in (11) where $Q(T, p, k - 1)$ is the correction factor and $Q(T, p, 0) = 1$.

$$p = \frac{\lambda}{T\mu} \quad (8)$$

$$p_0 = \left(\frac{T^T p^T}{T!(1-p)} + \sum_{j=0}^{T-1} \frac{T^j p^j}{j!} \right)^{-1} \quad (9)$$

$$Q(T, p, j) = \sum_{k=j}^{T-1} \frac{(T-j-1)!(T-k)T^k p^{k-j} p_0}{(k-j)!T!(1-p)}, \quad j = 0, \dots, T-1 \quad (10)$$

$$w_k = Q(T, p, k-1)(1-p)(p^{k-1}), \quad k = 1, \dots, T \quad (11)$$

4 The ε -constraint method

Several approaches exist for solving multi-objective problems such as weighted-sum, ε -constraint, and weighted-norm. The weighted-sum method, while popular, is not suitable for our problem because our solution space is integer and it is known that when the solution space is not convex, the weighted-sum method cannot find all solutions. However, both ε -constraint and weighted-norm approaches can find all solutions of integer problems. In this paper, we selected the ε -constraint method which was introduced by Haimes et al. (1971) and an extensive discussion can be found in Chankong and Haimes (1983). The idea of this technique is to minimize or maximize one objective while the other objectives are bounded at acceptable fixed values. If we have a multi-objective problem, the formulation of the ε -constraint method is given as follows (Ehrgott 2005).

The multi-objective problem	The ε -constraint problem
Minimize $[f_1(x), f_2(x), \dots, f_p(x)]$	Minimize $f_j(x)$
Subject to: $x \in X$.	Subject to: $f_k(x) \leq \varepsilon_k$ $k = 1, \dots, p; k \neq j$ $x \in X$ where $\varepsilon \in R^p$

We briefly discuss the concept of optimality as it relates to multi-objective problems. A feasible solution $\hat{x} \in X$ is called “efficient” or “Pareto optimal”, if there is no other $x \in X$ such that $f(x) \leq f(\hat{x})$. If $\hat{x}$ is efficient, the point $\hat{y} = f(\hat{x})$ is called non-dominated. A feasible solution $\hat{x} \in X$ is called weakly efficient or weakly Pareto optimal, if there is no other $x \in X$ such that $f(x) < f(\hat{x})$. If $\hat{x}$ is weakly efficient, the point $\hat{y} = f(\hat{x})$ is called weakly non-dominated (Ehrgott 2005). Thus solving a multi-objective problem results in a set of solutions, and the decision-maker should be interested in the Pareto set because it represents solutions that is better than any other with respect to at-least one of the criteria of interest.

To apply the ε -constraint approach to solve this problem, we have to reformulate the problem in the ε -constraint form. Since we have two criteria, efficiency (objective 1) and equity (objective 2) we have two choices. We choose the first alternative, which is to maximize objective 1 while objective 2 is bounded at an acceptable level for several reasons. Pragmatically, objective 1 aims to maximize coverage, which is the predominant measure used to rate the performance of EMS systems; and while being fair to rural regions is important, it is critical to provide coverage to as many calls as possible. Computationally, if we consider the range of values of the different objectives (Z_1 and Z_2), the second objective always has a smaller range, such that it is possible to explore every possible level Z_2 through the epsilon constraint method. Lastly, as discussed in Sect. 3.2, the formulation of the problem requires fewer constraints when objective Z_1 is used.

The details of the epsilon constraint method, in which we maximize Z_1 while the second objective is bounded at an acceptable value, ε_2 follow. Recall that we propose three alternatives to represent the second objective, resulting in are three bi-objective models, denoted as (a), (b), (c), depending on the objective chosen to be bounded (Z_{2a} , Z_{2b} , Z_{2c} , respectively). Model (a) is represented as (1) subject to (3)–(7) and (12), model (b) is represented as (1) subject to (3)–(7) and (13), and model (c) is represented as (1) subject to (3)–(7) and (14). The entire ε -constraint problem is represented as below.

$$\text{Maximize} \quad Z_1 = \sum_{i=1}^n \sum_{k=1}^{l_i} h_i w_k y_{ik} \quad (1)$$

$$\text{Subject to:} \quad (3)–(7)$$

$$\max_{i \in U} \left\{ \min_{j \in O} (d_{ij}) \right\} \leq \varepsilon_{2a} \quad (12)$$

$$\sum_{i=1}^n (1 - y_{i1}) r_i \leq \varepsilon_{2b} \quad (13)$$

$$\sum_{i=1}^n (1 - y_{i1}) \leq \varepsilon_{2c} \quad (14)$$

Where: ε_{2a} = the acceptable bound of objective Z_{2a}

ε_{2b} = the acceptable bound of objective Z_{2b}

ε_{2c} = the acceptable bound of objective Z_{2c}

5 Case study

Our case study uses data from the Hanover Fire/EMS department, which is located in Hanover County, VA. The Hanover EMS department responds to 911 calls 24 hours a day and serves a county of 474 square miles, with a population nearing 100,000 individuals. Based on zoning, all locations within Hanover County are classified as either rural or urban. Therefore, a method is needed for classifying each location as either rural or urban. In this case study we used the zoning provided by the county in order to reflect the granularity of the locations within the decision context of allocating public resources, which results in a vector $\mathbf{r}$ used to assign a value of 1 to rural zones and 0 to others. In addition we consider variations on this vector $\mathbf{r}$, beyond the county's current definition, based on population density information captured from GIS data. This vector $\mathbf{r}$ could be generalized to classify a zone according to any demographic attribute that is significant to service providers or policy makers, e.g. socio-economic background.

We point out that there is no consensus on the definition of rural/urban, since the definition of rural is multifaceted, depending on population density, distance to an urban metro area, zoning, age of the population, occupations of the population, employment levels, and other factors (Miller and Luloff 1981; Hall et al. 2006). The US Department of Agriculture (USDA) identifies nine definitions based on those used by other governmental agencies (USDA 2007), and numerous other definitions and taxonomies exist (Hall et al. 2006). The factors used to define whether an area is rural or urban would not only affect the vector $\mathbf{r}$ but could significantly affect the model results. For example, Hart et al. (2005) note the importance of defining rural areas in a contiguous manner, since policy decisions are markedly different when rural areas are non-contiguous.

The definition of rurality depends on the scale and granularity of the area of interest. For example, in an application that considers the entire United States, it may be appropriate to classify an entire county as either rural or urban. Indeed, the rural definition based on Census places classifies the county as 100% rural. However, such a definition would not be useful for allocating public resources, since they are allocated at a more local level. Hart et al. (2005) note that the definition of rurality should capture the most relevant elements with respect to the policy decision at hand, and the definition of rurality used in this analysis aims to address this need.

The data are collected from the Fire/EMS department during 2007, which capture the life-threatening calls received during 2007. Instead of assuming each demand point is located in the middle of an area, we divided the coverage area into 175 distinct zones. In this way, we ensure that coverage is more accurate and that originating demand is represented realistically. Currently, there are $m = 16$ existing station facility locations for locating EMS ambulances. All demand zones and station locations are shown in Fig. 1, where each demand zone is classified as either rural or urban based on zoning information provided by the county. We see from Fig. 1 that there are 6 potential station locations in urban areas (more than 90% of calls originate in urban areas) and 10 potential station locations in rural areas. The number of requested calls is collected separately for each demand zone. Based on the current data, requested calls did not originate from all 175 zones. Therefore, we ignore the zones that have no demand and only consider the $n = 122$ zones in which demand existed in 2007.

The input data to the model are the number of the requested calls (or call volume) in each demand zone, the geographical coordinates of the 122 demand zones and 16 potential locations, and the demand zone classification, as either rural or urban. Following current Hanover standards, we use a response-time threshold of 9 minutes for the covering objective

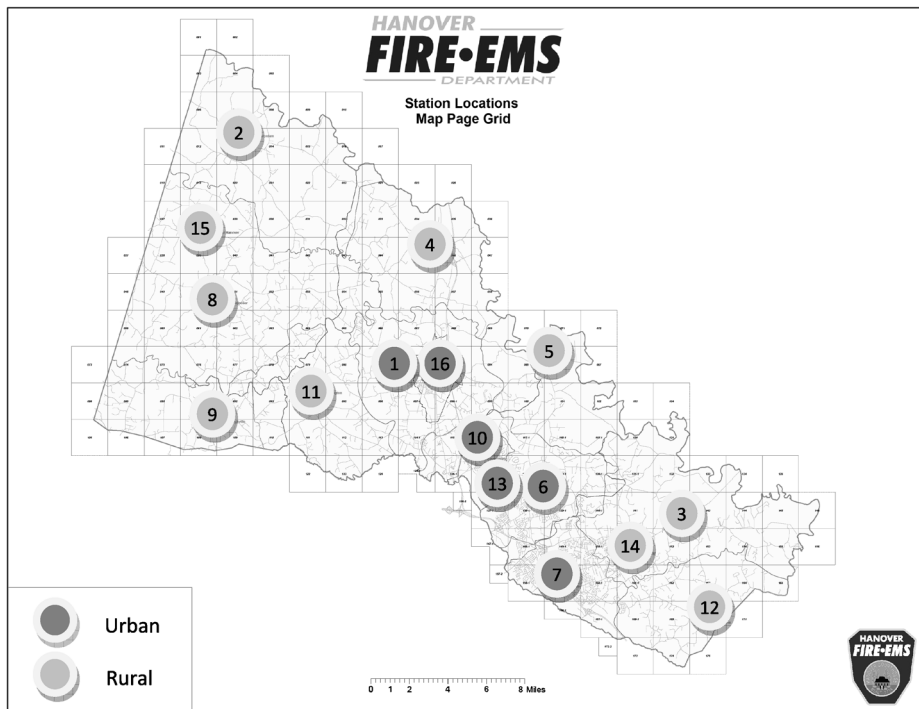


Fig. 1 Map of possible station locations and demand zones

(Z_1). Thus, a demand zone is said to be covered when there exists an EMS vehicle that is able to respond to a call in that zone within 9 minutes. In particular, we assume based on distance, average vehicle speed, and road conditions that for a call to be responded to within 9 minutes, at least one facility should be open within 4 miles of the demand zone. To set up the location of the station and demand zone, we drew grid lines over the area of interest, with one block representing 2 miles. The coordinates (a, b) of the stations and demand zones are used to calculate the distance between each demand zone and each station. Distance between two points can be measured in many ways (Drezner and Hamacher 2004). The most familiar two are rectilinear distance and Euclidean distance. In this case we use the Euclidean metric because approximately 70% of the Hanover County area is rural, and can thus be reached via highways or county roads. Given a demand zone i at (a_i, b_i) and a station location j at (a_j, b_j) , the distance (d_{ij}) between demand zone i and station j is calculated using the Euclidean metric. In this case, there are 1711 calls spread over 122 demand zones; given the set of possible station locations, there are 4 zones that cannot be covered, since they are more than 4 miles from the closest possible station. Therefore, the maximum percentage of coverage is 98.8%.

Based on the data during 2007, the average number of calls in Hanover is 1.2 calls/hour during the peak hours of operation when the call volume is essentially constant. The average service time per call is 74 minutes or 1.2 hours (note that this is not response time but also includes time in service). These data are used to compute the input parameters for our model.

6 Computational results

We use the data from the Hanover Fire/EMS department as described in Sect. 5, which is comprised of 122 demand zones and 16 possible station locations. We allow the total number of ambulances to be located in all stations to vary between 5 and 20, while the maximum number of ambulances that are allowed to be located at each station is 2.

6.1 Results of a single objective model

As a benchmark, we first consider the results using a single objective of maximizing the expected number of calls that are covered (Z_1). A single objective problem which consists of the Z_1 subject to constraints (3)–(7) was solved using the optimization software ILOG OPL 5.5 on a Dell Latitude D410 machine with Intel Pentium processor 1.73 GHz, 1 GB of RAM. The number of vehicles is varied from 5 to 20, which results in a total 15 cases. The run time was between 2 and 5 seconds per case. For clarity, the objective function value is rescaled by the total number of calls to reflect the proportion of calls that are covered, as shown in Table 1. As we increase the number of ambulances, the probability of a randomly selected vehicle being busy decreases, and the number of calls that are covered increases. Using a single objective, at least two ambulances are always located at station 1 and 6. These stations are located in urban areas and can serve a number of high call volume demand zones nearby. As more ambulances become available, stations 3, 4, 11, 14 and 15 are the next likely to be selected because they located near urban areas. Conversely, stations 2, 5 and 12, located in remote areas, are only selected when the number of available ambulances is high, or when coverage is already high.

When there are five ambulances, this single objective covering location model locates four ambulances in urban areas and one vehicle in a rural area, thus the majority of the uncovered demand zones are rural.

6.2 Results of the bi-objective model

To reduce the disparity between service in urban and rural areas, we would like to provide decision makers with more alternatives. Thus, we construct a bi-objective model which not only considers maximizing the expected number of calls that can be covered but simultaneously improves fairness to rural patients. As discussed in Sect. 4, we solve this problem using the ε -constraint method by first formulating the problem in the ε -constraint form. To this end, we need to find the bounds on ε_{2a} , ε_{2b} , and ε_{2c} . To do this, we solve the problem (3)–(7) with objective Z_1 (1) to optimality and obtain upper bounds on ε_{2a} , ε_{2b} , and ε_{2c} by calculating the values of Z_{2a} , Z_{2b} , and Z_{2c} after reaching optimality. To obtain lower bounds on ε_{2a} , ε_{2b} , ε_{2c} , we solve the problem (3)–(7) with objectives Z_{2a} , Z_{2b} , and Z_{2c} (2a), (2b), and (2c) one at a time.

Detailed results are provided for the case with 5 ambulances though similar conclusions hold for the case of more ambulances; furthermore, a discussion is later provided regarding the effects of increasing the number of ambulances. With 5 ambulances to be located, we find the maximum expected demand that can be covered is 1260.3 calls (73.66% of calls), and the upper bounds of the maximum distance between uncovered demand zones and their closest opened stations (ε_{2a}), the number of uncovered rural demand zones (ε_{2b}) and the number of uncovered demand zones (ε_{2c}) are 18, 63 and 69, respectively. To get the solution points, we solve the problem by maximizing the first objective while varying the value of ε_2 . Since we have three alternatives for the second objective, we have three models to solve.

Table 1 Result of single-objective problem which maximize covered demand Z_1

Number of ambulances	Probability vehicle will be busy	Expected demand that is covered (calls)	Opened stations (stations in bold face are located in rural areas)	Number of ambulances at each station	Coverage percentage
5	0.296	1260.3	{1 6 14 }	{2 2 1}	73.66
6	0.247	1365.0	{1 6 14 15 }	{2 2 1 1}	79.78
7	0.211	1448.4	{1 4 6 14 15 }	{2 1 2 1 1}	84.65
8	0.185	1508.1	{1 4 6 7 13 14 15 }	{2 1 1 1 1 1 1}	88.14
9	0.164	1558.5	{1 4 6 7 11 13 14 15 }	{2 1 1 1 1 1 1 1}	91.09
10	0.148	1589.3	{1 3 4 6 7 9 11 13 15 }	{2 1 1 1 1 1 1 1 1}	92.89
11	0.135	1614.6	{1 3 4 5 6 7 9 11 15 }	{2 1 1 1 2 1 1 1 1}	94.37
12	0.123	1636.5	{1 2 3 4 5 6 8 9 11 14 }	{2 1 1 1 1 2 1 1 1 1}	95.65
13	0.114	1648.9	{1 2 3 4 5 6 7 8 9 11 13 14 }	{2 1 1 1 1 1 1 1 1 1 1}	96.37
14	0.106	1659.4	{1 2 3 4 5 6 7 8 9 11 12 15 }	{2 1 1 1 1 2 1 1 1 1 1}	96.98
15	0.099	1668.0	{1 2 3 4 5 6 7 8 9 11 12 15 }	{2 1 1 2 1 2 1 1 1 1 1}	97.49
16	0.093	1675.3	{1 2 3 4 5 6 7 8 9 11 12 13 14 15 }	{2 1 1 2 1 1 1 1 1 1 1 1}	97.91
17	0.087	1679.6	{1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 }	{2 1 1 2 1 1 1 1 1 1 1 1 1}	98.16
18	0.082	1682.7	{1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 }	{2 1 1 2 1 1 1 1 1 1 1 2 1 1 1}	98.35
19	0.078	1684.9	{1 2 3 4 5 6 7 8 9 10 11 12 14 15 }	{2 1 1 2 2 2 1 1 1 1 2 1 1 1}	98.47
20	0.074	1686.8	{1 2 3 4 5 6 7 8 9 10 11 12 14 15 }	{2 1 1 2 2 2 1 1 2 1 2 1 1 1}	98.59

Fig. 2 Solution points of model (a)—second objective is to minimize the distance between uncovered demand and opened stations, with *solid circles* representing non-dominated solutions

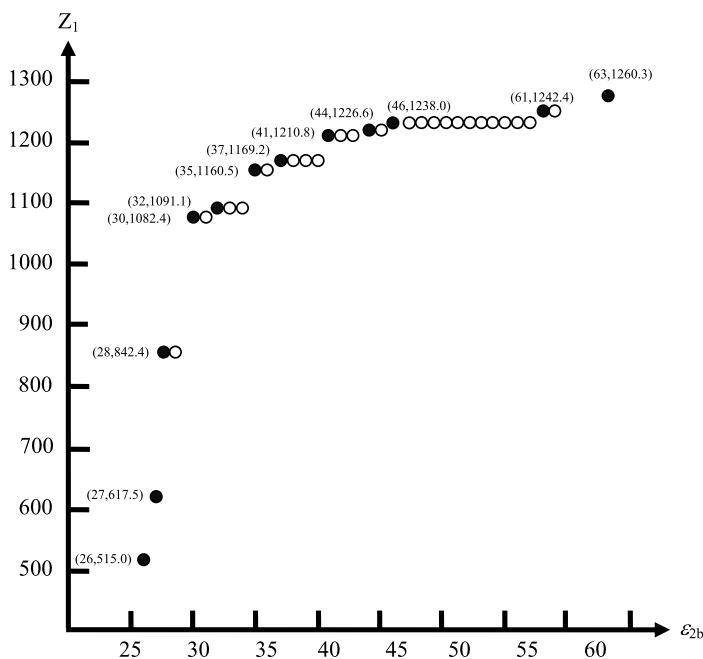
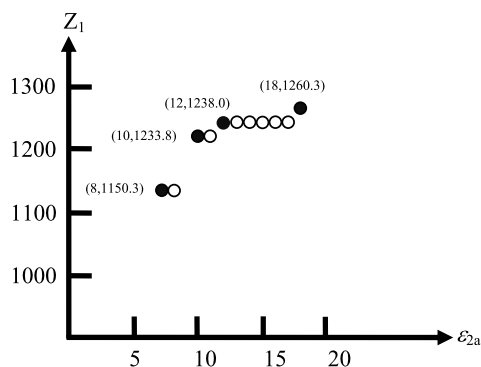


Fig. 3 Solution points of model (b)—second objective is to minimize the number of uncovered rural demand zones, with *solid circles* representing non-dominated solutions

Figures 2, 3, 4 show all the solution points that are found by maximizing the first objective while decreasing the values of ε_{2a} , ε_{2b} , and ε_{2c} from their upper bounds down to their lower bounds, in increments of 1. The feasible range of epsilon values are [8, 18], [26, 63], and [36, 69], resulting in 83 sub-problems. These problems were solved using the optimization software ILOG OPL 5.5 on a Dell Latitude D410 machine with Intel Pentium processor 1.73 GHz, 1 GB of RAM, the run time was between 1 and 2 seconds per sub-problem. In Figs. 2–4, an open circle represents a solution point of the ε -constraint method; a solid circle represents a non-dominated solution.

When Z_{2a} is selected, we build model (a) by selecting Z_{2a} to be the second objective. From Fig. 2, we see that for Z_{2a} values between 8 and 18 miles result in maximal coverage

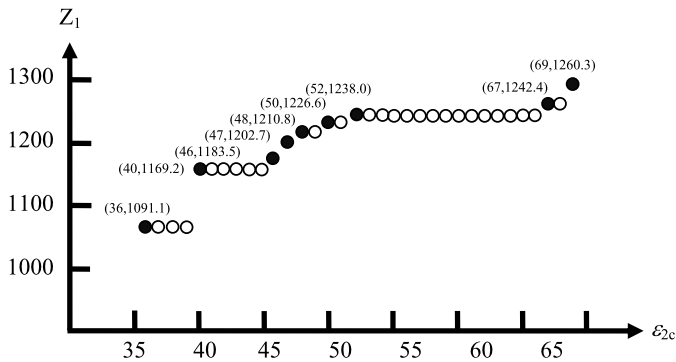


Fig. 4 Solution points of model (c)—second objective is to minimize the number of uncovered demand zones, with *solid circles* representing non-dominated solutions

ranging between 1150.3 and 1260.3 calls. Similarly, we build models (b) and (c) by selecting Z_{2b} and Z_{2c} as the second objective, respectively. Figure 3 shows all solution points for model (b) which minimizes the number of uncovered rural demand zones as a secondary objective. From Fig. 3, we see that for second objective values in the range of 26 and 63 zones the resulting second objective values are between 515.0 and 1260.3 calls. Figure 4 shows all solution points for model (c) which minimizes the number of uncovered demand zones; when the second objective values are between 36 and 69 zones the resulting first objective values are between 1090.1 and 1260.3 calls. For all three models we see that the best first objective value Z_1 is reached at the maximum ε_2 value, and if we decrease the second objective function value by decreasing ε_2 , the Z_1 objective function value deteriorates. Recall that all the solution points found are weakly non-dominated points, though solid circles are the non-dominated solutions. All three models contain the single objective solution as part of the dominated set with $Z_1 = 1260.3$. The other solutions “improve” the issue of fairness by sacrificing coverage.

The values of the objective functions and the corresponding non-dominated solution sets of models (a)–(c) are shown in Tables 2–4, respectively. In general, as we improve fairness, more stations are opened in rural areas, decreasing disparity of service at the expense of reducing coverage in urban areas. Note that the use of model (a) tends to locate ambulances at the most remote stations (i.e., those near the edges of the county) to reduce the maximum distance between uncovered demand zones and closest opened stations. Station 2 is the most isolated station, and the results of model (a) indicate that when station 2 is open we cannot achieve coverage higher than 67% ($Z_1 \leq 1150.3$) (this is not obvious from Table 2 since it shows only non-dominated solutions), however as we allow the maximum distance between uncovered demand zones and the closest open station (Z_{2a}) to increase then more urban stations are included in order to maximize expected coverage and the remote stations drop out of the solution set, since these have low call volumes. Also, model (a) yields a small solution (efficient solution) set. This implies that we can achieve close to maximal expected coverage without adversely affecting the distance between uncovered zones and open stations when the minimum maximum distance between uncovered zones and open stations is not required to be too small. For example, in Fig. 2 we see that to achieve maximal coverage of 1260.3, the distance is 18 miles but we can reduce this distance by 33% (to 12 miles) while only reducing coverage by 1.8% (to 1238.0).

Model (b) also opens stations in rural areas, in contrast to the stations chosen in model (a), stations are opened in order to cover as many as rural demand zones as possible, independent

of how far away uncovered stations may be from opened stations. Thus, model (b) opens rural stations having relatively high demands. Model (c) minimizes the total number of uncovered demand zones, in either rural or urban areas. Thus model (c) opens more urban stations than models (a) or (b). In contrast to the single objective model, the decision maker may choose to sacrifice call volume to cover a larger geographical region. For example, instead of the single objective solution which places one ambulance at one rural station (and leaves 63 rural zones uncovered) for an expected coverage of 1260.3, the decision maker may choose to reduce the expected coverage to 1210.8 calls but leave only 41 rural stations uncovered; in other words, coverage may be decreased by only 4% while zones covered are increased by 35%. We also note that, models (b) and (c) tend to “split” ambulances, rather than pairing ambulances at the same station. This is desirable when wanting to increase the number of covered zones, though it may be undesirable in the sense that it reduces backup coverage in high demand areas. For many decision makers a large solution set might be desirable, as it represents more options to choose from. In this case, Model (b) provides the largest set of efficient solutions.

In order to provide a guideline for decision makers on selecting an efficient solution generated from these three bi-objective models, we show the solution points of the three models in terms of the trade-off between coverage and fairness (e.g. the percent decrease in coverage as a result of some percent reduction in each disparity measure) in Figs. 5a, 5b, 5c.

Fig. 5a Coverage (Z_1) & fairness (Z_{2a}) trade-off based on the upper bound of Z_1 and the lower bound of Z_{2a} from model (a)—second objective is to minimize the distance between uncovered demand and opened stations (solution numbers correspond to solutions in Table 2)

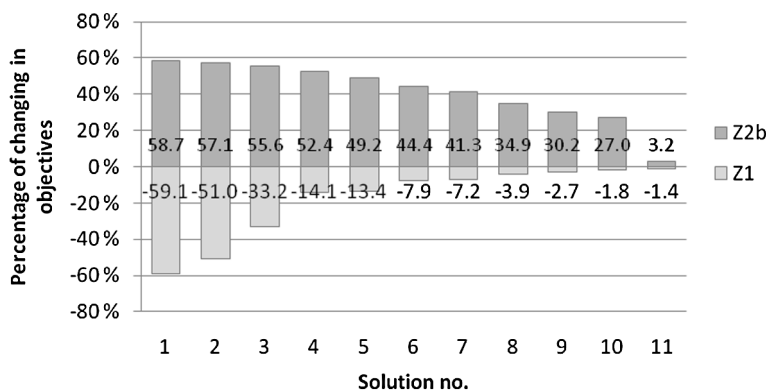
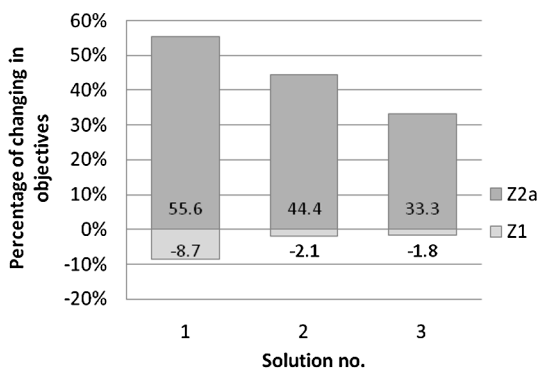


Fig. 5b Coverage (Z_1) & fairness (Z_{2b}) trade-off based on the upper bound of Z_1 and the lower bound of Z_{2b} from model (b)—second objective is to minimize the number of uncovered rural demand zones (solution numbers correspond to solutions in Table 3)

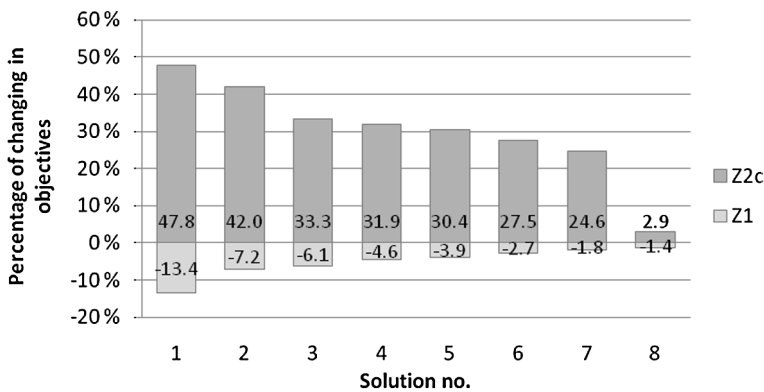


Fig. 5c Coverage (Z_1) & fairness (Z_{2c}) trade-off based on the upper bound of Z_1 and the lower bound of Z_{2c} from model (c)—second objective is to minimize the number of uncovered demand zones (solution numbers correspond to solutions in Table 4)

The single objective covering problem, with no consideration of fairness is considered to be a baseline (0%), this achieves a maximum coverage of 1260.3 calls. By reducing coverage by a small percentage, we are able to improve quality of service in rural areas. For example, from Fig. 5a we see that a 1.8% reduction in coverage is offset by a 33.3% reduction the maximum travel distance to a customer. Similarly from Figs. 5b and 5c we see that a 1.8% reduction in coverage can be offset by a 27.0% in the number of rural zones and a 24.6% increase in total zones covered, respectively.

6.3 Sensitivity analysis to rural classification

In this section, we would like to provide a discussion on how zone classification impacts the solution, especially for model (b) whose objective directly depends on zone classification, as either urban or rural. As we mentioned before the 175-zone classification is provided by the county, there are 137 rural zones and 41 urban zones where all the urban zones are located contiguously in the middle of the county (see Fig. 7a). In order to test the sensitivity of our results, we use different criteria to classify a zone. To do this we use different population density cut-offs to classify zones as urban or rural, with the exception that small dense areas are not considered urban unless they are contiguous to other urban area. The GIS data regarding county population is obtained from the U.S. Census Bureau in TIGER (Topologically Integrated Geographic Encoding and Referencing) file format (U.S. Census Bureau 2000). Arc GIS 10 is selected to manage the data. The station map provided by the county is digitized and incorporated into the census map, and then the attribute data of these two sources are joined. We classify the county area into 8 classes based on population density using quantile-defined intervals as displayed in Fig. 6. As mentioned previously, the urban area is contiguously located in the middle of the county where population density is high. In this section we consider three different rural classifications, namely, rural-1, rural-2, and rural-3 in order to see how this impacts the solution. Rural-1 is the original rural/urban zoning provided by the county while rural-2 and rural-3 are modified versions of the original zoning obtained by reducing and expanding the area of urban zones, respectively. In rural-2 zoning, only areas with population density higher than the 6th quantile, or at least 566 persons per sq. mile, are classified as urban zones. While in rural-3 zoning, areas with population density higher than the 4th quantile, or at least 93 persons per sq. mile, are classified as urban zones.

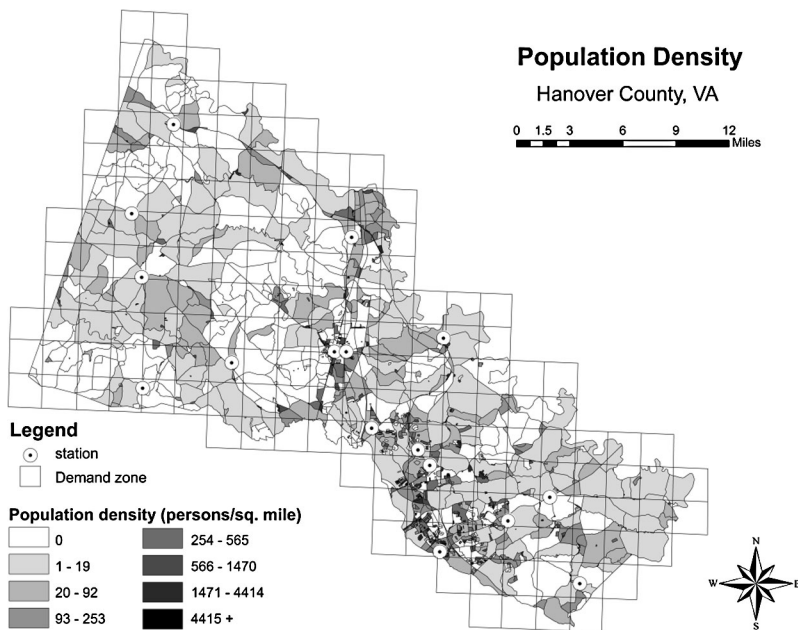


Fig. 6 Map of demand zones displaying population density

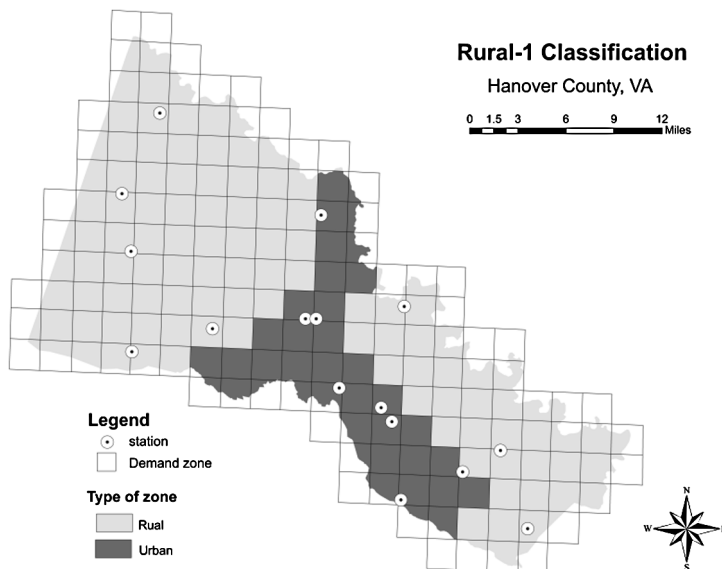


Fig. 7a Map of demand zones with rural-1 classification

Note that this results in 153 rural zones in rural-2 zoning and 123 rural zones in rural-3 zoning. Maps of how the 175-zones are designated with different rural classifications; rural-1, rural-2, rural-3, are shown in Figs. 7a, 7b, 7c.

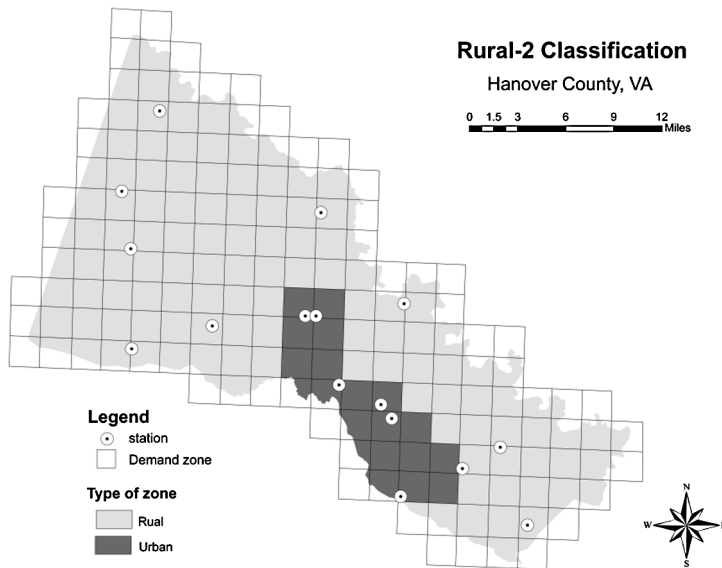


Fig. 7b Map of demand zones with rural-2 classification

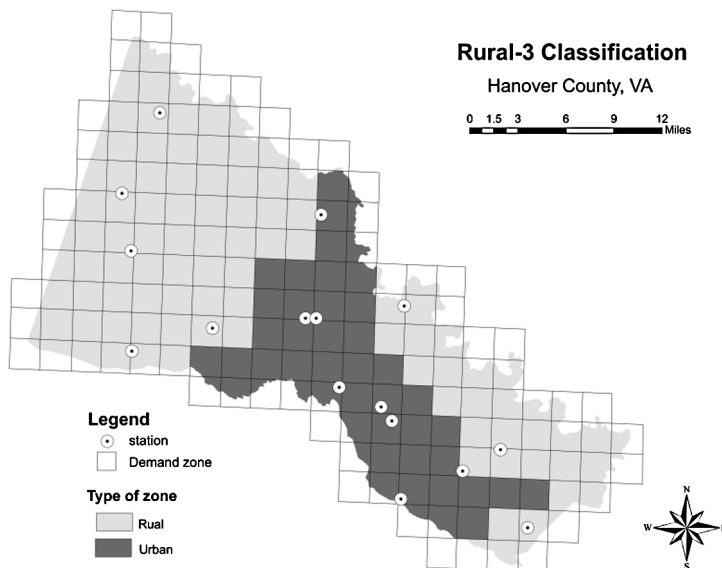


Fig. 7c Map of demand zones with rural-3 classification

We again solve the three bi-objective models with the three different rural zone classifications. Note that the solution set of model (b) is the only one that is affected by this, (though the discussion of trade-offs is affected for all three models). Figure 8 shows non-dominated solution sets of model (b). As the number of rural zones changes, this affects the feasible range of ε_2 which in turn results in more or fewer of solution points. Using rural-2 classifi-

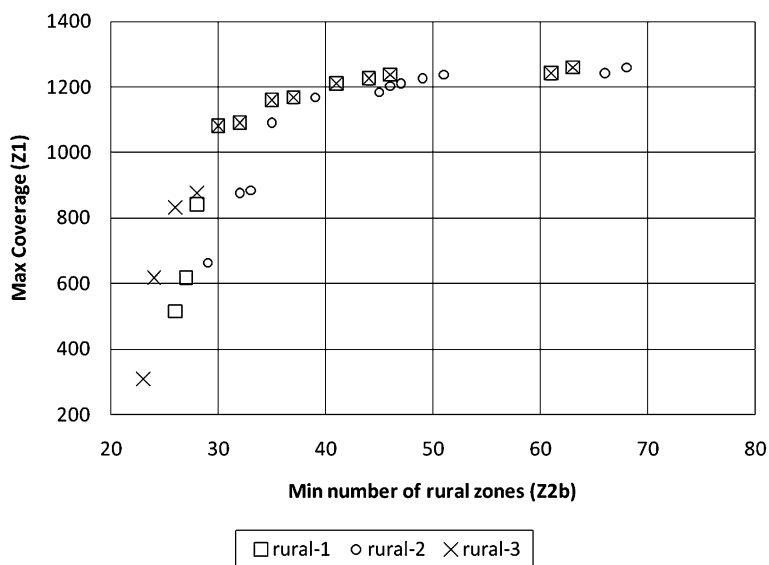


Fig. 8 Solution points of model (b) with rural-1, rural-2, and rural-3 classification areas

cation, where the number of rural zones is increased, we are able to cover at most 29 rural zones, while using rural-1 and rural-3 classifications we are able to cover at most 26 and 23 rural zones, respectively. Moreover, the optimal set of stations to open is also changed when zoning is changed. In this example, more than 50% of solutions to model (b) with rural-2 classification and rural-3 classification are the same as the solution points of model (b) with rural-1 classification. With different classifications, these three models are still able to reach the maximum coverage and keep the same pattern of the non-dominated curve.

While model (b) directly improves rural disparity by covering more rural zones, its solution depends on how the system administrator classifies zones. On the other hand, models (a) and (c) that improve rural disparity through proxies (by reducing the maximum traveled distance and number of uncovered zones, respectively) result in solutions that are not sensitive to zoning classifications. Model (a) improves service to patients who live the farthest away from an open station; these patients typically live in rural zones. Model (c) tries to increase number of covered zones, whether urban or rural; using this as a secondary objective however usually results in urban zones being added to the coverage set first as they have higher population density and therefore higher call volume. Therefore the mechanisms by which each of the three models reduces rural disparities are different. A discussion on how to choose the “best” model is provided in the next section.

7 Further discussion and post-analysis of solutions

We see in the previous section that the three bi-objective models we proposed help to reduce disparities between rural and urban areas. However, for the decision maker to decide which model is most appropriate, she must be able to evaluate these three models with respect to some common metrics. To compare the three models head-to-head, in Figs. 9a, 9b, 9c, we plot the proportion of rural/urban zones of those covered for all efficient solution points generated by each bi-objective model. Note that a zone can be covered by more than one

Fig. 9a Number of covered rural/urban zones obtained from model (a)—second objective is to minimize the distance between uncovered demand and opened stations

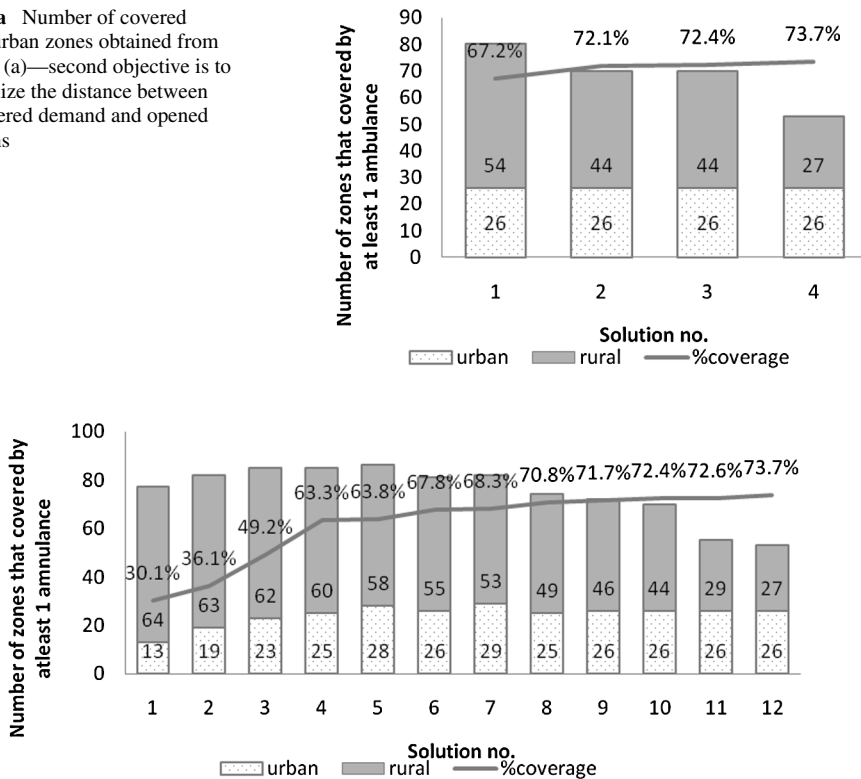


Fig. 9b Number of covered rural/urban zones obtained from model (b)—second objective is to minimize the number of uncovered rural demand zones

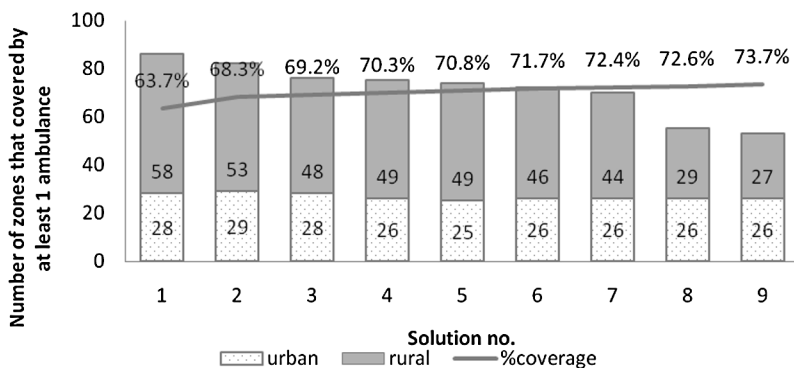


Fig. 9c Number of covered rural/urban zones obtained from model (c)—second objective is to minimize the number of uncovered demand zones

ambulance; here we say a zone is covered if it is covered by at least one ambulance. Since model (b) has the objective of increasing the number of rural zones covered, this allows us to see how well models (a) and (c) perform as proxies without directly relying on the zone classification scheme. With a single objective of maximizing coverage 54 zones are covered,

whose composition is 26 urban zones and 27 rural zones. In Fig. 9a, we see that model (a) with objective Z_{2a} improves rural disparity by covering more rural zones without sacrificing coverage of urban zones. Note that model (a) improve rural disparity by trying not to have multiple coverage in urban zones. For model (b) the Z_{2b} objective directly forces increased coverage to rural zones, its solutions improve by increasing rural coverage while decreasing urban coverage. For model (c) the Z_{2c} objective focuses on reducing number of uncovered zones, so it improves rural disparity by increasing the number of covered zones overall. Note that the total number of covered zones in model (b) decreases in some cases to improve rural disparity while the total number of covered zones in model (c) always increase to improve rural disparity.

Clearly, using any of the bi-objective models proposed in this paper provides more alternatives to the decision makers. However, comparing these solutions is difficult. Another way to compare these three solutions is to evaluate them with respect to other metrics. In practice, if we cannot introduce specific metrics directly within the objective function, providing performance metrics for post-analysis evaluation is useful for comparing different portfolios of ambulance locations. Thus, for each model (using the data from the case study with the original r vector, or rural-1 classification), we include the result of all three secondary objectives, as well as the average distance from uncovered zones to the closest open station (which captures the average distance between demand zones and the closest open station), and the weighted average distance from uncovered zones to the closest open station (which captures the average distance an ambulance can expect to travel per call). These latter two measures are reported in Tables 2, 3 and 4 for models (a), (b) and (c), respectively. Note that these metrics could not be included directly as objective functions in our models because they can only be formulated as nonlinear functions.

If we use the average distance or the weighted average distance from an uncovered zone to its closest station as supplementary criteria, the results from using model (a) may be preferred to the results from using models (b) and (c); we achieve the same average distance but achieve higher coverage simply by opening a different set of stations. That is, even though more rural zones are uncovered, these uncovered zones are located closer to open stations, improving the chance that they will receive service in a reasonable time. This is important since response time is directly linked to survivability (McLay and Mayorga 2010).

The results shown here are for the case of five available ambulances. With a single objective the best, we can achieve is 73.66% coverage with 69 uncovered zones and an average distance of 10 miles from uncovered demand zones to open stations. Using a bi-objective model, we can reduce the average distance from uncovered zones to open stations by sacrificing coverage. The only way to improve both metrics (decrease the average distance between uncovered demand zones and open stations and increase the number of calls that can be covered), is to increase the number of ambulances to be located. Increasing the number of ambulances improves all metrics. Furthermore the reductions in disparities that can be achieved by using a bi-objective model in comparison to a single objective are reduced. This is not necessarily because the model is less efficient but rather because there are enough ambulances to improve service for all regions. Therefore, the proposed bi-objective model may be most useful when there are fewer available ambulances, when identifying fair ambulance location portfolios is most difficult.

8 Conclusions

Traditional covering location models can lead to solutions that result in disparities in service between different demographic or geographic regions. Optimally locating ambulances

Table 2 The objective values and the non-dominated solutions of model (a)

Solution no.	Expected demands covered (Z_1)	Maximum distance from uncovered zone to closest station (Z_{2a})	Opened stations {rural; urban} {stations in bold face have 2 vehicles located}	Number of uncovered rural zones	Number of uncovered zones	Number of uncovered demands	Average distance from uncovered zone to closest station	Weighted average distance from uncovered zone to closest station	SDEV
1	1150.3	8	{2 8 14; 6 1}	36	42	196	6.5	6.3	1.00
2	1233.8	10	{8 14; 6 1}	44	50	206	7.0	6.6	1.40
3	1238.0	12	{14 15; 1 6}	46	52	219	7.2	6.9	1.77
4	1260.3	18	{14; 1 6}	63	69	296	10.0	9.0	3.66

Table 3 The objective values and the non-dominated solutions of model (b)

Solution no.	Expected demands covered (Z_1)	Number of uncovered rural zones (Z_{2b})	Opened stations {rural; urban} {stations in bold face have 2 vehicles located}	Maximum distance from uncovered zone to closest station	Number of uncovered zones	Number of uncovered demands	Average distance from uncovered zone to closest station	Weighted average distance from uncovered zone to closest station	SDEV
1	515.0	26	{2 3 8 9; 1}	10	45	982	6.6	6.7	1.14
2	617.5	27	{2 3 8 11; 13}	12	40	838	6.9	6.4	1.65
3	842.4	28	{3 9 15; 1 13}	10	37	529	6.4	6.0	1.02
4	1082.4	30	{3 9 15; 1 6}	10	37	186	6.4	6.1	1.05
5	1091.1	32	{3 11 15; 1 6}	10	36	180	6.6	6.3	1.32
6	1160.5	35	{9 14 15; 1 6}	10	41	180	6.5	6.2	1.08
7	1169.2	37	{11 14 15; 1 6}	10	40	174	6.7	6.4	1.32
8	1210.8	41	{3 15; 1 6}	12	48	225	7.3	6.8	1.81
9	1226.6	45	{14 15; 1 6 13}	12	50	200	7.3	7.0	1.80
10	1238.0	46	{14 15; 1 6}	12	52	219	7.2	6.9	1.77
11	1242.4	61	{14; 1 6 13}	18	67	277	10.1	9.2	3.67
12	1260.3	63	{14; 1 6}	18	69	296	10.0	9.0	3.66

Table 4 The objective values and the non-dominated solutions of model (c)

Solution no.	Expected demands covered (Z_1)	Number of uncovered zones (Z_{2c})	Opened stations {rural; urban} {stations in bold face have 2 vehicles located}	Maximum distance from uncovered zone to closest station	Number of uncovered rural zones	Number of uncovered demands	Average distance from uncovered zone to closest station	Weighted average distance from uncovered zone to closest station	SDEV
1	1090.1	36	{3 11 15; 1 6}	10	32	180	6.6	6.3	1.32
2	1169.2	40	{11 14 15; 1 6}	10	30	174	6.7	6.4	1.32
3	1183.5	46	{4 14 15; 1 6}	12	42	152	7.1	7.2	1.84
4	1202.7	47	{3 7 15; 1 6}	12	41	208	7.3	7.0	1.80
5	1210.8	48	{3 15; 1 6}	12	41	225	7.3	6.8	1.80
6	1226.6	51	{14 15; 1 6 13}	12	44	200	7.3	7.0	1.80
7	1238.0	52	{14 15; 1 6}	12	46	219	7.2	6.9	1.77
8	1242.4	67	{13 14; 1 6}	18	61	277	10.1	9.2	3.67
9	1260.3	69	{14; 1 6}	18	63	296	10.0	9.0	3.66

to improve fairness is an important issue, and this paper proposes a bi-objective model to address this problem. In particular, we applied the ε -constraint method to solve a bi-objective covering location problem. The first objective is to maximize the number of requested calls that can be covered by the ambulances within a response time standard, the second objective is aimed at reducing disparity in service between rural and urban citizens. The second objective is modeled in three ways: to (a) minimize the maximum distance between uncovered demand zones and opened stations or to (b) minimize the number of uncovered rural demand zones or to (c) minimize the number of uncovered demand zones

The results are obtained using data from Hanover County, a rural/suburban county in Virginia. In addition, the results depend on a single method for classifying demand zones as either rural or urban. As mentioned in the Introduction, there are numerous methods for classifying demand zones, some of which may significantly affect the results. We note that the method for classifying demand nodes as rural or urban should be selected with care while keeping the purpose of allocating EMS resources in mind. The results, therefore, should not be interpreted to provide a general policy for all types of EMS systems, since the results depend on travel distances and call locations that may not be characteristic of other urban and suburban areas. However, the proposed model can be used to reduce disparities in service for other types of EMS systems.

With one objective, we can only obtain the solution which maximizes number of requested calls that can be covered or we can obtain the solution which minimizes one of the secondary objectives. By using a bi-objective model, we can find all the solution points in between the best value of the first objective and the best value of the second objective. The solution points we find provide a set of efficient (non-dominated) solutions, or alternatives, that are very useful for decision makers wishing to take into account issues of fairness when locating EMS ambulances. While each model yields a set of non-dominated solutions that are not directly comparable, we propose two performance metrics to use as selection criteria: the average distance or the weighted average distance from an uncovered zone to its closest station. Under these criteria, model (a) which minimizes the maximum distance between uncovered zones and its closest open station always provides a better solution, without significant sacrifice to the first objective, though model (b) offers a larger Pareto set and therefore more options to the decision maker. However, the sensitivity analysis showed that model (b) is sensitive to zone classification while model (a) and model (c) are not. The final choice depends on the decision makers. The largest reduction in disparities is achieved when service is poor to average. This is important because it has been observed that levels of care are typically not as good in rural areas as compared to urban areas. Thus mediocre service in urban locations could translate to very poor service in rural areas; our model helps to provide solutions that mitigate these issues of fairness.

Analyzing models which consider issues of fairness in the delivery of EMS service is important. This paper proposes three bi-objective models to reduce disparities in service received by rural and urban citizens. We believe considering each fairness objective separately (resulting in three bi-objective models), as opposed to a multi-objective model which considers 4 objectives simultaneously is a valid approach for two reasons. First, the goal is to balance two conflicting criteria, efficiency and fairness. The three proposed fairness objectives have not been tested before; thus, it is important to test the effect of each proposed objective on the resulting solution (set of open stations). Second, we wish to help policy-makers draw insights into how to reduce disparities in service among different geographic regions, thus the solution should be easy to interpret. Extending this approach to take into account more than two conflicting criteria (e.g. including workload), or including criteria which may be easier to interpret (but may result in non-linear formulations) are important future areas of research.

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